

Navigating the gold rush

Of influencer economies and FOSS

Hi there, I'm Divya Mohan!

- Open source advocate & maintainer
- Former systems nerd
- Feline enthusiast



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What is a gold rush?

“A **gold rush** or **gold fever** is a discovery of **gold**—sometimes accompanied by other **precious metals** and **rare-earth minerals**—that brings an onrush of miners seeking their fortune.”

— Wikipedia



Source: Gold miners using the hydraulic mining method to find gold in Nevada County by Lawrence & Houseworth, 1866, via Library of Congress, Washington DC

So, what's the big deal?

The project shows how the gold rush laid waste to the forests of the Sierras and the oak and redwood forests surrounding San Francisco, home to grizzly bears. The once bountiful wetlands and meadows of the Sacramento and San Joaquin Valleys that had been home to huge herds of antelope, elk, pelicans, quail, and otters are now barren wastelands that need massive injections of water and chemicals to produce crops.

A total of 12 billion tons of earth, eight times the amount of land dug up for the Panama Canal, was dumped into the local rivers. In 1880 alone, 40,000 acres of farmland were destroyed and 270,000 acres severely damaged. The cleanup bill for the Iron Mountain mine alone has been estimated by the EPA at \$3 million to \$4 million a year—for 3,000 years.

“... this widespread mining has contributed to serious public health, safety, and environmental concerns. For example, the Environmental Protection Agency (EPA) estimates that abandoned hardrock mines have contributed to the contamination of 40% of the country's rivers and 50% of all lakes.”

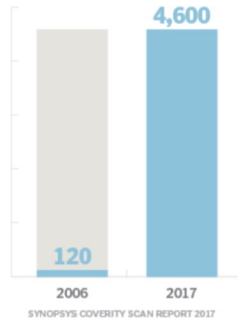


How does this relate to open source?

Open source by the numbers

Open source software usage among enterprises has exploded in the last decade, according to industry research. In the last three years, that usage has given way to open source contributions by enterprise developers, often under the auspices of corporate open source programs.

Number of open source projects on the rise

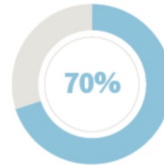


180,000
Number of open source projects
tallied in 2019 by IntroBooks
THE RISE OF OPEN SOURCE SOFTWARE E-BOOK

Open source everywhere



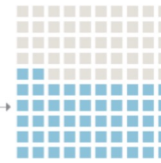
of codebases audited by security firm Synopsys in 2019 contained open source components



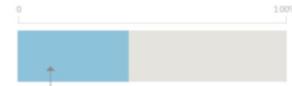
Percentage of open source within audited codebases

2020 OPEN SOURCE SECURITY AND RISK ANALYSIS REPORT

Benefits of open source contribution



52% of 2,700 participants in a Linux Foundation survey have open source programs or plan to create one



41% say open source programs are responsible for ensuring high code quality and frequent releases to open source communities

LINUX FOUNDATION TODO GROUP OPEN SOURCE PROGRAM MANAGEMENT SURVEY 2019

- **2023 saw the largest number of first-time open source contributors.** We continue to see commercially backed open source projects capture the largest share of first-time contributors and overall contributions—but this year, we also saw generative AI projects enter the top 10 most popular projects for first-time contributors. We're also seeing notable growth in private projects on GitHub, which increased 38% year over year and account for more than 80% of all activity on GitHub.

Why this? Why now?

Enter the trifecta

The internet

The rise of viral
marketing a.k.a.
Word of mouse

Influencer marketing

Who/what is an influencer?

Influencers are defined as “people who built a large network of followers, and are regarded as trusted tastemakers in one or several niches”

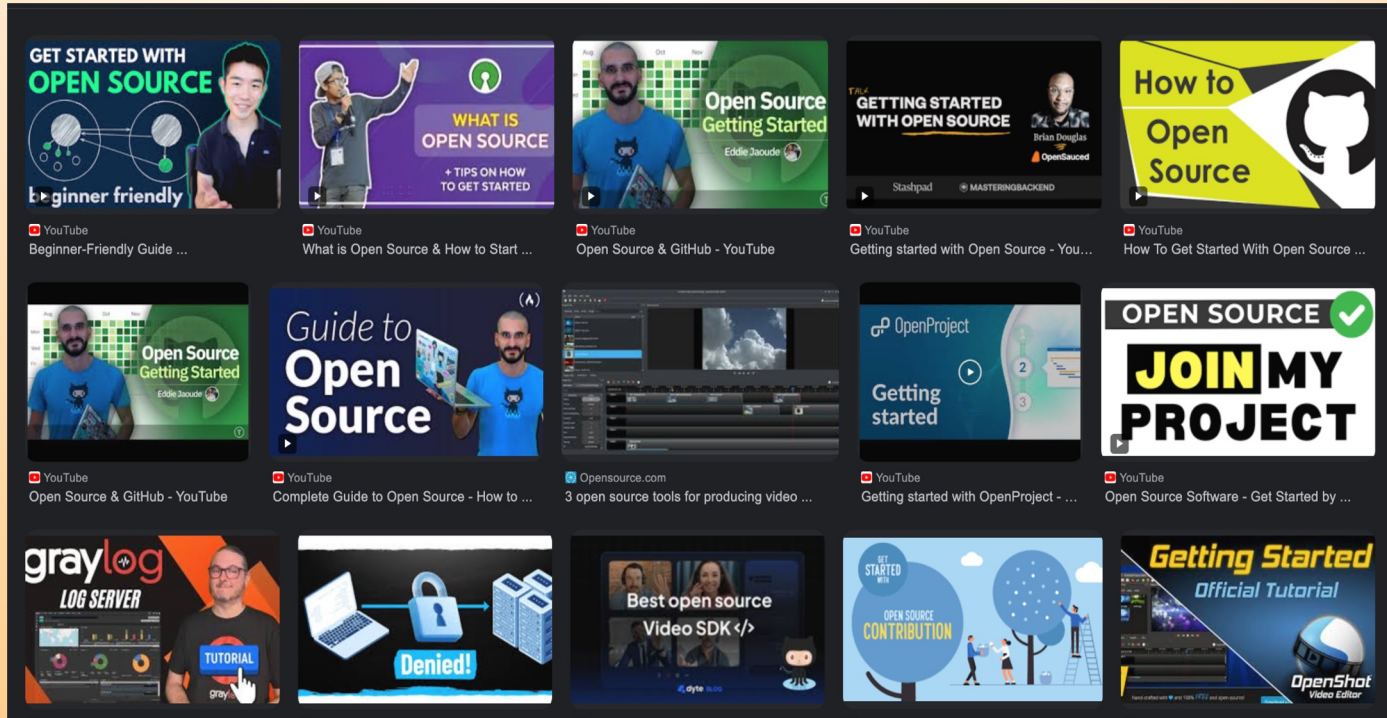
What “can” they achieve?

loyalty toward social media

influencers leads followers to have a strong sense of product/brand association and reality by perceiving influencers' product/brand and advertisements as more persuasive and authentic [4]. Accordingly, users are more likely to behave favorably for the product/brand social media influencers advertise in order to endorse the influencers and support the influencers' fame and social status [7].

Isn't that kinda good, though? 🤔

What the first page looks like today



#TBT Hacktoberfest 2020

This was k/website in October 2020

☐		reverting hacktoberfest changes	approved	cncf-cla: yes	lgtm	sig/docs	size/S		7
#24360 by jimangel was merged on Oct 4, 2020									
☐		Update README-hi.md	language/hi	sig/docs	size/XS				6
#24359 by NityanDesai was closed on Oct 3, 2020									
☐		Update README-hi.md	cncf-cla: no	sig/docs	size/XS				4
#24358 by NityanDesai was closed on Oct 3, 2020									
☐		Spell check	cncf-cla: no	sig/docs	size/XS				9
#24357 by Sushant012 was closed on Oct 3, 2020									
☐		Update README-de.md	cncf-cla: no	sig/docs	size/S				7
#24354 by shaikhahad18 was closed on Oct 3, 2020									
☐		Update ingress.md version	cncf-cla: no	language/en	sig/docs	size/XS			7
#24352 by artisticman was closed on Oct 3, 2020									
☐		Hacktoberfest readme update	approved	cncf-cla: yes	lgtm	sig/docs	size/S		4
#24342 by jimangel was merged on Oct 2, 2020									
☐		apostrophe	cncf-cla: no	sig/docs	size/XS				6
#24338 by coderajat24 was closed on Oct 2, 2020									
☐		Update README-de.md	cncf-cla: no	sig/docs	size/XS				6
#24317 by harshp1732 was closed on Oct 1, 2020									
☐		improved docs	cncf-cla: no	sig/docs	size/XS				6
#24313 by simransharmarajni was closed on Oct 1, 2020									
☐		Update README-hi.md	cncf-cla: no	sig/docs	size/XS				6
#24310 by Akatsuki-123 was closed on Oct 1, 2020									
☐		Update linkchecker.py	cncf-cla: no	sig/docs	size/XS				6
#24309 by karan97144 was closed on Oct 1, 2020									

☐		Create 404.html	cncf-cla: no	sig/docs	size/S				
#24278 by adityakumar48 was closed on Oct 1, 2020									
☐		Translation in english	cncf-cla: no	sig/docs	size/XS				
#24275 by ass-coder was closed on Oct 1, 2020									
☐		Update LICENSE	cncf-cla: no	sig/docs	size/S				
#24273 by akanshujamal was closed on Oct 1, 2020									
☐		Thank you 😊, Added	cncf-cla: no	sig/docs	size/XS				
#24272 by Alexjr977 was closed on Oct 1, 2020									
☐		Updated Readme	cncf-cla: no	sig/docs	size/S				
#24268 by meetpatel125 was closed on Oct 1, 2020									
☐		Improved Docs	cncf-cla: no	sig/docs	size/XS				
#24267 by rvdhiver was closed on Oct 1, 2020									
☐		Imporved The Docs	cncf-cla: no	sig/docs	size/XS				
#24264 by zeeshanthedev590 was closed on Oct 1, 2020									
☐		Create Website	cncf-cla: no	sig/docs	size/XS				
#24260 by arjunkendr-e was closed on Oct 1, 2020									

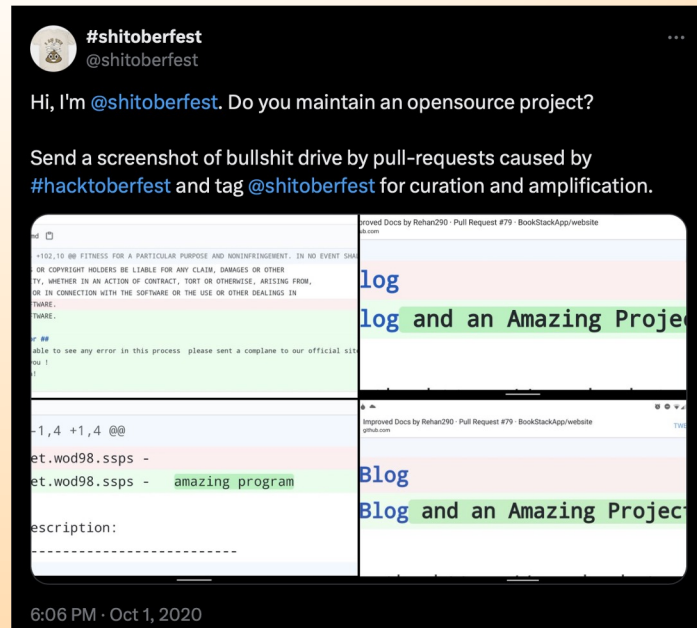
We weren't the only ones affected...



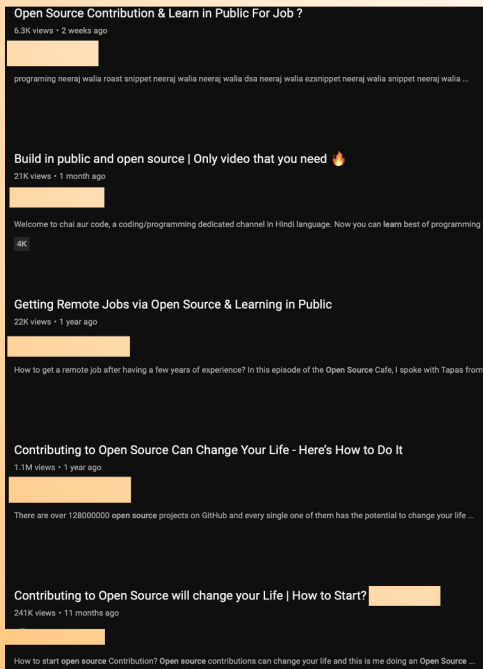
DigitalOcean's Hacktoberfest is Hurting Open Source

September 30, 2020 | [Tweet](#) | posted in [open source](#)

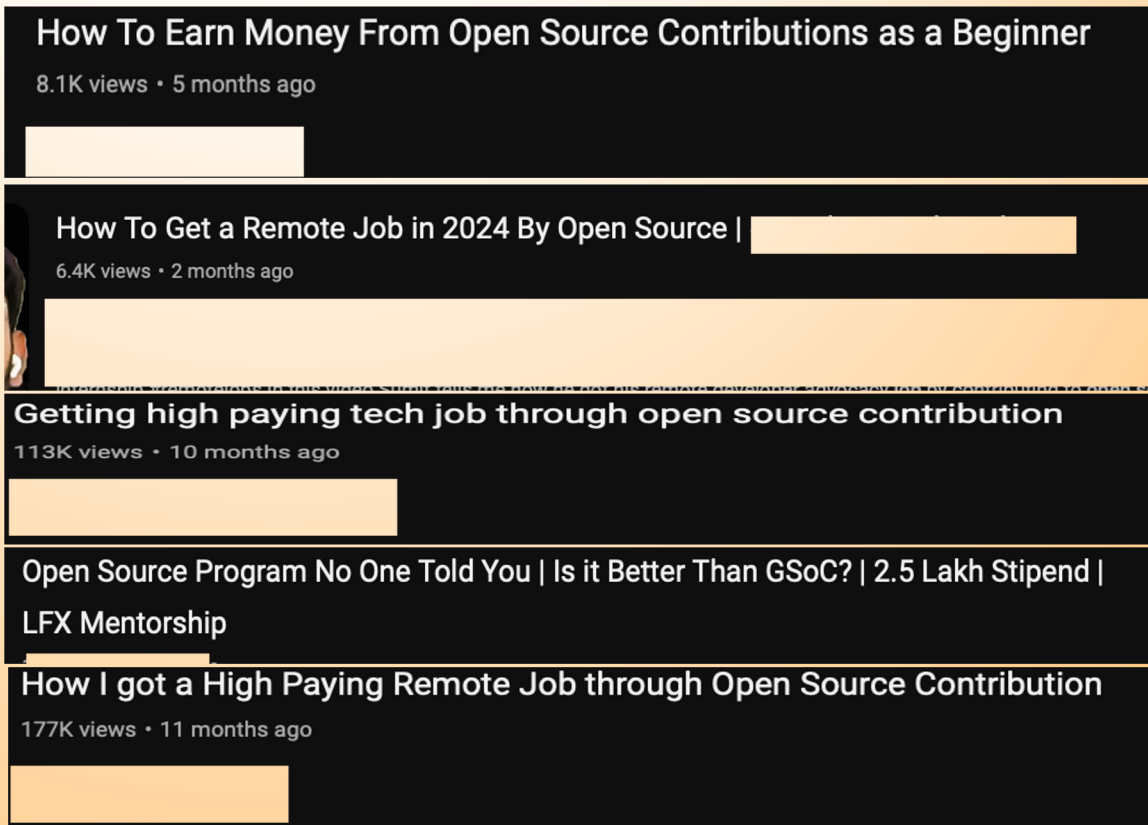
@Divya_Mohan02



And then, it got worse...



@Divya_Mohan02



And then...it got worse



Github is Your New Resume

github | git | organizations | people

GitHub Isn't Just a Git Hosting Service Anymore

Now, it's becoming a social network for developers, and that's great

No, I really do mean it!

Making Your Github Green Again

 obsessive compulsive Order · Follow

3 min read · Jul 26, 2018

fake-it-til-you-make-it

Have you come across to someone that thinks you don't deserve a job because you don't have GitHub contributions?

Run this program.

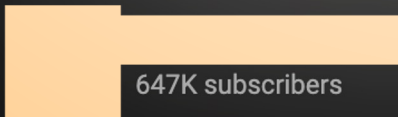
Usage

For example, commit 10 commits max every day for two years, and avoid committing on Saturday and Sunday:

```
go run main.go -max=10 -skip-weekend=true
```



Hack Github Contribution Graph in 20 seconds 🕶️



647K subscribers

Subscribe

Isn't all publicity good for business?

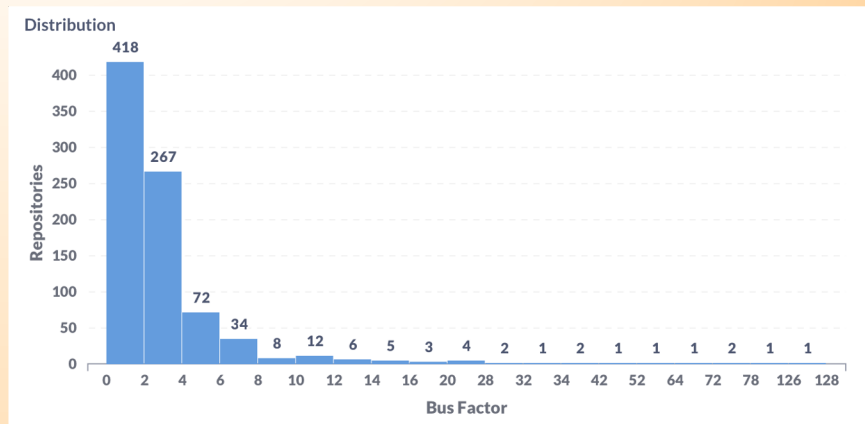
Nope!

The best and most obvious way to judge an open-source project is to look at the code but this can be kind of tedious and sometimes you don't like what you see there, so an alternative that we have all naturally developed on our own or have been advised to, is to see how many people have starred a project, and then pick the one with the most stars.

Venture investors are “hardwired” to look for fast growth in startups seeking investment, says Pratima Aiyagari, a partner at venture firm Nauta Capital. Open source projects can operate for years without generating significant revenue, she says, so investors look for other growth signals, of which GitHub stars are just one. The success of companies like business software firm Mulesoft and collaborative software development platform Gitlab has drawn strong interest into open source companies, she says. “VC money has been pouring into the space.”

Nope!

- Bus factor for most popular open source projects found to be <2 .
- High risk of OSS projects going unmaintained if main contributors a.k.a. maintainers aren't active.



Two-thirds of maintainers are making an effort to create a welcoming atmosphere around their projects

Across all the projects you maintain, which of the following efforts are being made to strengthen community and project long-term health? (Choose all that apply)

Creating a welcoming atmosphere for all identities	67%
Efforts to improve the inclusivity of information and communication (i.e. getting rid of technical jargon and avoiding words that express bias)	37%
Community onboarding and outreach	35%
Mentorship opportunities (e.g., Outreachy and Google Summer of Code)	28%
Tools for non-developer contributions	27%
Efforts to improve the accessibility of information and communication (i.e. for the visually impaired or non-English speakers)	21%
Efforts to address project burnout	20%
Creating/using a conflict resolution process	14%
None of the above	17%

Maintainer burnout is real. Almost 60% of maintainers have quit or considered quitting maintaining one of their projects

How do we fix this?

The communication pillar

A few ideas...

- Communicate better!
 - Better indicators and metrics for open source advocacy
 - What are we measuring?
 - Process?
 - Output?
 - Outcome?
 - Are these formalized?
 - Have these been adopted?
 - Have these efforts been streamlined?

Metric basics

- **Metric title:** Advocacy/Outreach
- **Metric summary** (1-2 sentences): Advocacy and outreach initiatives are important non-code contributions toward helping spread awareness about the project. Not only do they help engage potential contributors to open source, but they also significantly help with the adoption/use of the project contributing to the health and sustainability of the community.
- **Why should this metric model be created?** (1-2 sentences): Although this is partially covered under [Types of Contributions](#) and [Community Awareness](#), advocacy efforts (and their effectiveness) aren't measured in yet AFAIK. It's a quintessential chicken-and-egg problem. Since we don't know what to measure, there is a need for indicators to define contextual progress with respect to the efforts employed, and there are very few formalised/streamlined programs within projects championing advocacy efforts. Not only are these efforts fragmented, but they are also heavily reliant on the bandwidth of maintainers/leaders of the project, who can't measure/sponsor these efforts without proper indicators of success.

The governance pillar

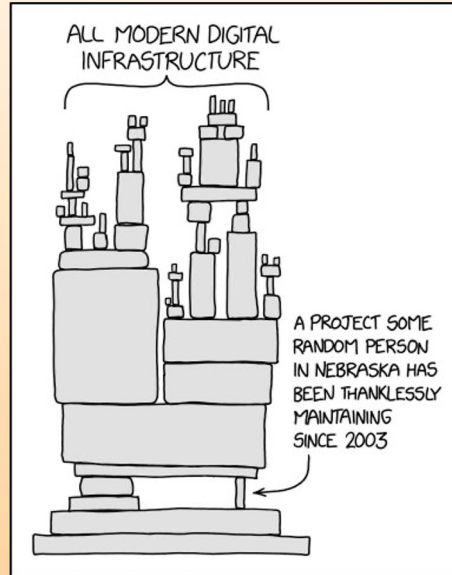
A few ideas...

- Better DEI initiatives & metrics to welcome a broader audience
 - Nurturing and growing paid third-party mentorship programs such as Outreachy, Season of Docs, Summer of Code, LFX mentorship etc.
 - Pair mentoring efforts with maintainers
 - Making DEI a first order citizen in your project
 - More representation in leadership
 - Committees/subprojects for DEI within OSS projects

The maintainer pillar

A few ideas...

- Devise better ways of ensuring maintainer sustainability
 - Let's not make this meme a standard!



A few ideas...

- Around 75% of OSS maintainers also double up as super contributors
 - Make open source “viable” for OSS maintainers by paying them
 - Helps them focus on maintaining projects
 - Enables them to build strong contributing pipelines via
 - Formal mentorship programs
 - Formal recruitment/new contributor onboarding programs

Resources

- [Open Source Maintainers, The Linux Foundation](#)
- [The xkcd comic on dependency](#)
- [Advocacy/messaging metric](#)
- [The CHAOSS project](#)
- [Open source maintainers, Tidelift survey](#)
- [Bus factors of top GitHub projects, Metabase](#)
- [The GitHub Black Market That Helps Coders Cheat the Popularity Contest, Kari McMahon, Wired](#)

Resources

- [How much are GitHub stars worth to you?](#)
- [Kim J, Kim M. Rise of Social Media Influencers as a New Marketing Channel: Focusing on the Roles of Psychological Well-Being and Perceived Social Responsibility among Consumers. Int J Environ Res Public Health. 2022 Feb 18;19\(4\):2362. doi: 10.3390/ijerph1](#)
- [De Veirman et al., 2016, p. 1](#)
- [State of Octoverse report, 2023](#)

Resources

- Synopsys report from 2019
- [From Gold Rush to Rot – The Lasting Environmental Costs and Financial Liabilities of Hardrock Mining](#)
- [Found SF](#)

I am on the internets.



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<https://linkedin.com/in/divya-mohan0209>



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Thank you!